

UK Public Spending | Power BI Project Overview

Project Objective

This project analyses UK government departmental expenditure using HM Treasury Public Expenditure Statistical Analyses data. The dashboard was designed to transform complex public spending data into an accessible executive-level reporting tool, allowing users to assess overall expenditure, departmental allocations, budget composition and changes in spending over time.

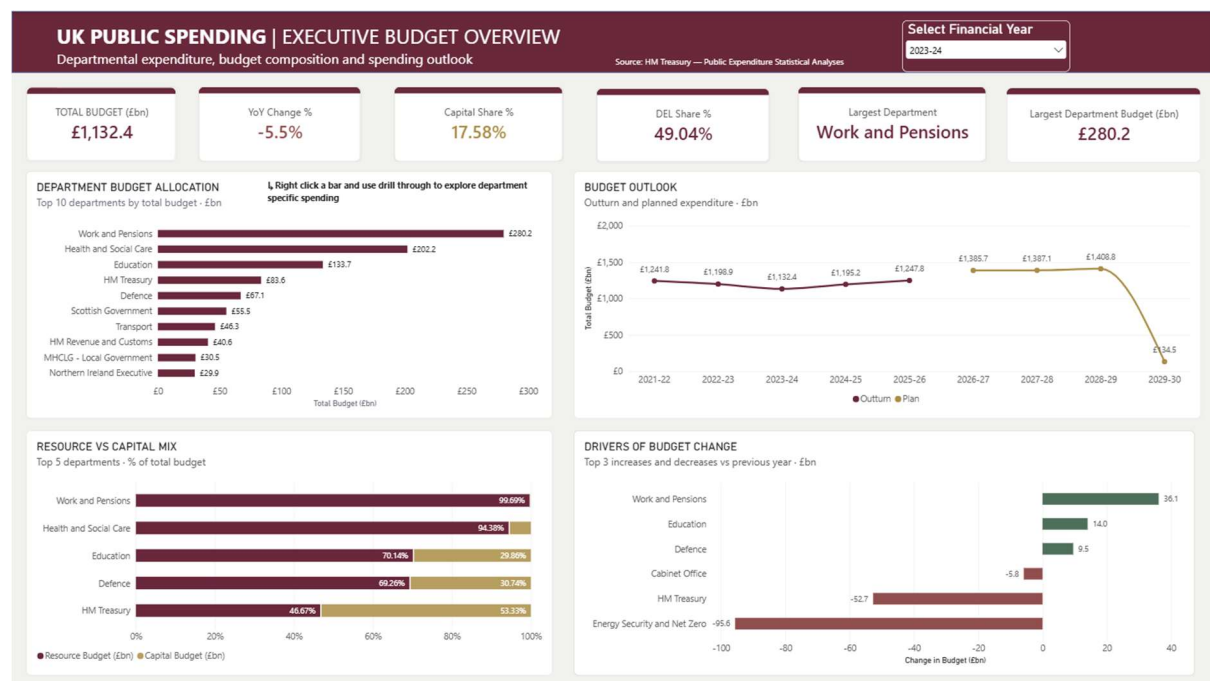
The report combines a high-level **Executive Budget Overview** with an interactive **Department Deep Dive**, enabling users to move from the overall fiscal position into the detailed spending structure of individual government departments.

Tools & Skills:

Power BI · Power Query · DAX · Data Modelling · Data Transformation · Drillthrough · Interactive Filtering · Financial Data Visualisation

DASHBOARD 1 - Executive Budget Overview

The Executive Budget Overview provides a high-level assessment of UK departmental expenditure for a selected financial year. It is designed to quickly communicate the **scale, distribution, composition and movement of government spending**.



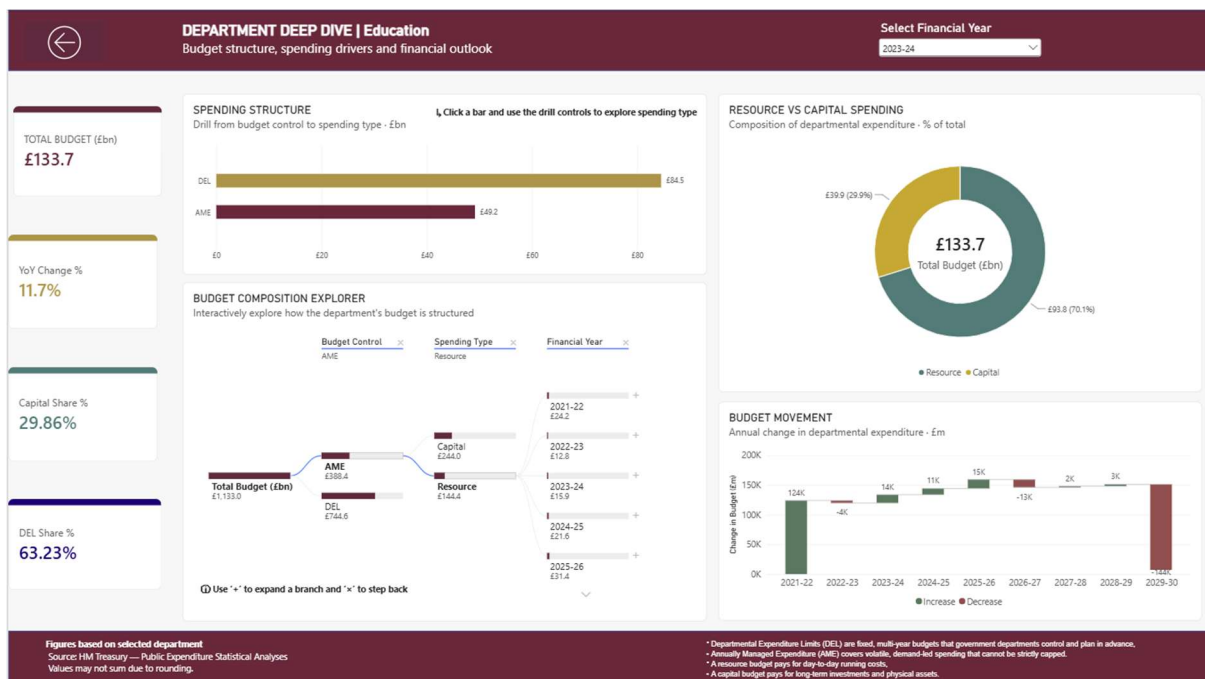
- **Executive KPIs** summarise total budget, year-on-year change, capital share, DEL share and the largest spending department.
- **Department Budget Allocation** ranks the ten largest departments by total expenditure and provides drill through access to individual departments.
- **Budget Outlook** distinguishes historical outturn expenditure from planned future spending, providing a forward-looking view of the public spending profile.

- **Resource vs Capital Mix** compares day-to-day expenditure with longer-term capital investment across the largest departments.
- **Drivers of Budget Change** identifies the departments responsible for the largest increases and decreases in expenditure compared with the previous year.
- A **dynamic financial-year selector** allows the entire executive view to be analysed across different reporting periods.

The dashboard allows decision-makers to identify **where government spending is concentrated, which departments are driving changes in expenditure, and how the balance between resource and capital spending differs across government.** Combining historical outturns with planned expenditure also provides context on how the overall spending profile is expected to develop.

DASHBOARD 2 - Department Deep Dive

The Department Deep Dive allows users to move beyond the headline figures and investigate the **structure, composition and evolution of expenditure within an individual government department.**



Users can access the page directly through drill through from the Executive Budget Overview, maintaining the selected department and financial-year context.

- **Department-specific KPIs** show total budget, year-on-year change, capital share and DEL share.
- **Spending Structure** separates expenditure between Departmental Expenditure Limits (DEL) and Annually Managed Expenditure (AME), with further drill-down into spending type.
- **Resource vs Capital Spending** shows the proportion allocated to day-to-day expenditure compared with investment in longer-term assets.
- The interactive **Budget Composition Explorer** uses a decomposition tree to investigate how expenditure is distributed across budget control, spending type and financial year.
- **Budget Movement** tracks annual changes in departmental expenditure, highlighting periods of increasing and decreasing spending.
- Synced filters and drillthrough functionality create a consistent analytical journey between the executive and departmental views.

Technical Implementation

The project required the transformation and restructuring of HM Treasury expenditure data in **Power Query**, followed by the creation of a structured Power BI data model. **DAX measures** were used to calculate dynamic KPIs including total expenditure, year-on-year changes, capital shares and DEL shares.

Interactive functionality was implemented through **drillthrough, synced slicers, hierarchical drill-down and decomposition-tree analysis**, allowing the report to function as an analytical tool rather than a static dashboard.

Data Source

HM Treasury — Public Expenditure Statistical Analyses (PESA)

Figures may not sum exactly due to rounding.

Full interactive PBIX file available on request.